Answer: A

14 $$\frac{GM_{sun}M_{satellite}}{R_{orbit}^{2}} - \frac{GM_{earth}M_{satellite}}{d_{between\ earth\ and\ satellite}^{2}} = M_{satellie}\left(\frac{4\pi^2}{T_{satellite}^{2}}\right)R_{orbit}$$

The resultant gravitational force on the satellite = gravitational force by the Sun – gravitational force by the Earth. Hence even if radius of orbit is lesser than that of the Earth, the period is the same.

Answer: D

15 The increase in PE = $\left(\frac{-GM_{Earth}M}{\quad}\right) - \left(\frac{-GM_{Earth}M}{\quad}\right)$